

1. Administrative & Study Identification

**Full Study Title:** A First-in-human, Multi-center, Randomized, Participant- and Investigator-blinded, Placebo Controlled, Single Ascending Dose Study to Assess the Safety, Tolerability, and Pharmacokinetics of EDK060 in Adults With Charcot-Marie Tooth Type 1A (CMT1A) **Short Title/Acronym:** A First in Human Study of EDK060 in Adults With CMT1A **Protocol Number:** CEDK060A12101 **NCT Number:** NCT07140614 **Sponsor Name:** Novartis (Novartis Pharmaceuticals Canada, Inc. / Novartis AG) **Investigational Product:** EDK060 (RNA-interfering technology targeting PMP22) **Phase of Development:** Phase 1 **Medical Monitor:** Novartis Clinical Operations Lead [Placeholder]

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To characterize the safety, tolerability, and pharmacokinetics of EDK060 as compared to placebo in adult patients with CMT1A. **Secondary Objectives:** To evaluate the incidence of Adverse Events (AEs) and Serious Adverse Events (SAEs) by treatment group, including changes in vital signs, laboratory measures, neurological examinations, and electrophysiology measures qualifying as AEs from baseline up to day 140.

**2.2 Background & Rationale** To address the lack of disease-modifying therapies for Charcot-Marie-Tooth Disease Type 1A (CMT1A). EDK060 aims to directly modify disease progression and potentially halt nerve damage by targeting the root cause of the condition—the overexpression of the PMP22 gene—rather than solely alleviating symptoms. This First-in-Human study represents a crucial step in evaluating this novel RNAi-based technology.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Participant- and Investigator-blinded) **Randomization:** Randomized

**3.2 Planned Enrollment & Duration Target \$N\$:** 28 participants **Number of Sites:** Multi-center **Estimated Study Start:** 30-Sep-2025 **Estimated Primary Completion:** 26-Feb-2028

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Male or female, 18 to 60 (inclusive) years of age at the time of screening. 2. Clinical diagnosis of CMT1A, including verified documentation of genetic testing showing duplication of the PMP22 gene by an accredited laboratory. 3. Detectable upper extremity nerve conduction velocities (sensory and/or motor) in at least one extremity at screening, CMTNSv2R score >2 and ≤20\$, and muscle weakness evidenced by foot dorsiflexion MRC grade <5 in at least one limb.

**4.2 Exclusion Criteria** 1. Pregnant or nursing female participants; or women of child-bearing potential not using highly effective contraception. 2. History of cardiac, renal, liver, hematological, or immune system disorders. 3. Use of any drug intended to modify the course of CMT1A within 6 months from screening (e.g., PTX-3003) or history of compression neuropathy/ other conditions causing peripheral neuropathy within the last 6 months.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Single Ascending Dose (EDK060 Dose A, Dose B, Dose C, Dose D, or matching placebo). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Single dose administration followed by a follow-up observation period.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard symptomatic therapies that do not alter the disease progression. **Prohibited:** Any drug intended to modify the course of CMT1A within 6 months prior to screening (including but not limited to PTX-3003: baclofen, sorbitol, and naltrexone in combination).

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## 6. Study Timeline & Visit Schedule

| Visit ID          | Window (Days) | Key Activities/Procedures                                                                                                  |
|-------------------|---------------|----------------------------------------------------------------------------------------------------------------------------|
| Screening         | Up to Day 0   | Informed Consent, genetic verification, clinical/neurological evaluation, verification of upper extremity nerve conduction |
| Baseline          | Day 0         | Randomization, Single IV Infusion of EDK060 or Placebo                                                                     |
| Interim Follow-up | Up to Day 140 | PK sampling, serial skin biopsies, repeated venipuncture, safety monitoring, vital signs                                   |
| End of Study      | Day 140       | Final neurological and electrophysiology examinations, AE/SAE final collection, Exit Interview                             |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Overall characterization of the safety, tolerability, and pharmacokinetics of EDK060. **Measurement Method:** Clinical assessment of Adverse Events (AEs), Serious Adverse Events (SAEs), and laboratory analysis of blood/plasma concentrations to determine PK profiles.

**7.2 Secondary & Exploratory Endpoints** Incidence of AEs and SAEs categorized by treatment group up to day 140. Changes in vital signs, laboratory measures, neurological examinations, and electrophysiology measures that qualify and are reported as AEs.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous monitoring of vital signs, laboratory tests, and targeted evaluation of potential risks associated with serial skin biopsies and repeated venipuncture. **Serious Adverse Events (SAEs):** Reported within standard Phase 1 24-hour expedited timelines. **Data Monitoring Committee (DMC):** An independent safety review committee will evaluate ascending dose cohorts to determine progression to subsequent dose levels.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Safety Set (all participants who received at least one IV infusion) and PK Analysis Set. **Sample Size Justification:**  $N = 28$  is deemed sufficient to establish initial human safety parameters, maximum tolerated dose constraints, and PK profiles for the tested dose ranges. **Handling of Missing Data:** Dealt with per standard operating procedures for Phase 1 SAD studies, with no primary imputation applied to missing safety metrics.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Strict clinical oversight during the single infusion and intensive subsequent follow-ups to ensure protocol adherence. **Audit Trail:** Secure electronic data capture (EDC) implementation for eCRF locking and timely safety data query resolution.

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## 11. Study Identifiers & Governance

**Primary ID:** CEDK060A12101 **Registry Number:** NCT07140614 **Sponsor/Lead Organization:** Novartis **Study Title:** A First in Human Study to Assess the Safety, Tolerability, and Pharmacokinetics of EDK060 in Adults With CMT1A **Official Regulatory Title:** A First-in-human, Multi-center, Randomized, Participant- and Investigator-blinded, Placebo Controlled, Single Ascending Dose Study to Assess the Safety, Tolerability, and Pharmacokinetics of EDK060 in Adults With Charcot-Marie Tooth Type 1A (CMT1A)

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## 12. Clinical Framework (CMT1A Specific)

**Primary Condition:** Charcot-Marie-Tooth Disease, Type 1A (CMT1A) **Diagnosis Threshold:** Verified documentation of genetic testing showing duplication of the PMP22 gene **Medical Methodology:** RNA-interfering (RNAi) technology **Optimization Target:** Direct disease modification via reduction of PMP22 expression

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## 13. Study Procedures & Methodology

**Management Pathway:** Single Ascending Dose (SAD) study combined with comprehensive clinical outcome assessments **Education Protocol (HCP):** Site training for IV dosing procedures, accurate collection of serial skin biopsies, and electrophysiology tracking **Education Protocol (Patient):** Comprehensive onboarding detailing the necessity of repeated venipuncture and skin biopsies **Quality Audit Mechanism:** Phase 1 safety data reviews and centralized electrophysiology/biopsy reading processes

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Clinical monitoring up to at least 140 days post-infusion **Onsite Visit Cadence:** Intensive multi-day immediate post-infusion assessments, transitioning to scheduled follow-up visits **Remote Monitoring Frequency:** Between scheduled physical visits for general safety verification **Checklist Review Interval:** Neurological and electrophysiology follow-ups as determined by protocol timepoints

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## 15. Sign-off & Approval

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|---------------------------------|--------|-----------|---------------|------|------|------|-------------|
| Role                            | Name   | Signature | Date          | :--- | :--- | :--- | :---        |
| <b>Principal Investigator</b>   | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |             |
| <b>Clinical Operations Lead</b> | [Name] | _____     | [DD-MMM-YYYY] |      |      |      | <b>Lead</b> |
| <b>Statistician</b>             | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |             |